

Model development to predict organic acid accumulation over grape berry development

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Les modifications climatiques en cours affectent déjà et continueront dans le futur d'impacter la viticulture. L'élévation prévisible des températures (+2 à + 8°C d'ici à 2100) vont induire une augmentation du rapport sucres/acides dans les baies à la récolte, diminuant leur potentiel œnologique. L'interprofession viticole manque d'outils prédictifs permettant de prédire avec suffisamment de précision l'évolution du rapport sucres/acides au cours du développement des baies, en fonction de la température. Les modèles écophysiologiques ont le potentiel pour produire de tels outils. L'objectif de ce travail était donc développer un modèle mécaniste permettant de simuler l'évolution du contenu en acides organiques au cours de la croissance et de la maturation de la baie de raisin. Ce modèle est basé sur le processus de stockage vacuolaire des acides malique et tartrique par un processus de piégeage d'acides. Il a été adapté au raisin en partant de modèles développés pour prédire le contenu en malate dans les pêches et les bananes (Lobit *et al.*, 2006 ; Etienne *et al.*, 2014). Un modèle spécifique a également été développé pour l'accumulation du tartrate, et ces deux modèles ont été couplés à un troisième modèle permettant de prédire l'évolution du pH vacuolaire de cellules de la baie de raisin. Au final, le modèle global obtenu permet de prédire, en utilisant comme variables d'entrées la température ambiante et la teneur en potassium des baies les profils d'évolution des contenus des baies en acides malique et tartrique ainsi que du pH vacuolaire.

Current climate changes are already affecting and will continue in the future to impact viticulture. The expected rise in temperature (+2 to + 8 °C by 2100) will increase in the sugar/acid ratio in berries at harvest, reducing their oenological potential. The wine industry interprofession lacks predictive tools to predict with sufficient precision the evolution of the sugar/acid ratio during berries development in response to ambient temperature. Ecophysiological models have the potential to produce such tools. The objective of this work was to develop a process-based model to simulate the evolution of organic acid content during the growth and maturation of the grape berry. This model is based on the process of vacuolar storage of malic and tartaric acids by acid trap. It has been adapted to grapes from models developed to predict malate content in peaches and bananas (Lobit *et al.*, 2006; Etienne *et al.*, 2014). A specific model has also been developed for tartrate accumulation, and these two models have been coupled with a third model to predict evolution of the vacuolar pH of grape berry cells. In the end, the overall model obtained makes it possible to predict, by using as input variables the ambient temperature and the potassium content of the berries, the evolution profiles of the contents of the berries in malic and tartaric acids as well as the vacuolar pH.

Vine – modeling – organic acid – acidity – temperature

Laboratory presentation

My Internship took place in the INRA-University of Bordeaux-BSA joint unit 1287 “Ecophysiologie et Génomique Fonctionnelle de la Vigne” (EGFV, Dir. Dr N. Ollat (before the 1/4/2018 Pr. S. Delrot), 44 permanent employees) at the ISVV (Institute of Wine and Vine Sciences) on the INRA Bordeaux-Aquitaine center.

EGFV joint unit is constituted by researchers from Environment and Agronomy, Biology and Plant Breeding of INRA divisions and Professors and Assistant Professors from Bordeaux University and Bordeaux Science Agro. The Unit works on vine physiology and its response to environmental constraints through different approaches such as ecophysiology, molecular physiology, biochemistry and genetics. It is organized alongside two research axis: the rootstock-scion interaction (axis 1) and the grape berry quality (axis 2). My work takes place in axis 2 research lines, particularly the FRIMOUSS modeling project. Being a partner of Bordeaux PhD program, the unit hosts several PhD and master students.

UMR EGFV is the French leader in research dedicated to the interaction of vines with the environment, in the context of climate change since the initiation of work on this topic in 2007.

As examples, the Unit recently coordinated with INRA Montpellier (UMR Innovation) a national project of INRA's Metaprogramme "Adaptation of Agriculture and Forestry to Climate Change" (ACCAF metaprogramme). This LACCAGE project (Long Term Impacts and Adaptations to Climate Change in Viticulture and Enology) proposed a multidisciplinary and comprehensive approach to study the impacts of climate change and the long-term adaptation pathways of viticulture and production of wine in France (Project LACCAGE).

The UMR has been coordinator of a workpackage on Adaptation of viticulture to climate change within the European program KBBE Innovine (2013-2016). (<https://www6.bordeaux-aquitaine.inra.fr/egfv/Activites-scientifiques/Faits-marquants>)



Abbreviations list

Ca⁺⁺ : calcium

Mg⁺⁺ : magnesium

K⁺ : potassium

V-ATPase : vacuolar ATPase

V- PPiase : vacuolar pyrophosphatase

TCA : Krebs cycle

ALMT : aluminium activated malate transporter

DTC : mitochondrial dicarboxylate/ tricarboxylate channel

ATP : adenosine tri phosphate

ADP : adenosine di phosphate

ΔG_{ATP} : variation of free energy of ATP hydrolysis

$\Delta\Psi$: electromotive force

pH_{vac} : vacuolar pH

pH_{cyt} : cytosolic pH

R : perfect gaz constant

F : Faraday constant

T : temperature

RRMSE : Relative Root Mean Squared Error

RMSE : Root Mean Squared Error

RRMSEP : Relative Root Mean Squared Error of Prediction

cv : cultivar

QTL : quantitative trait loci

sd : standard error

°K : degrees Kelvin

°C : degrees celcius

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I Introduction

1.1 Context and general introduction

Grapevine is one of the most valuable fruit crops in the world, due to its various uses such as wine and liquor making. In France, vineyards occupy around 3% of arable land and produce between 40 and 55 millions of hectolitres of wine each year, accounting more than 800 000 jobs (FranceAgriMer). Wine quality depends on berry quality at harvest which is a complex trait. It results of asynchronous accumulation of primary (sugar, organic acids) and secondary metabolites (polyphenols, aromas...). Specific fruit composition at harvest is influenced by static and dynamic environmental factors, together defining its enological potential (Conde *et al.*, 2007). Particularly, the organic acid-to-sugar ratio is a key parameter for harvesting high quality grapes. This ratio, which is controlled by complex genotype x environment interactions will be affected by the ongoing climate change and particularly temperature which will increase it, leading to unbalanced, lower quality wines. Berry acidity is determined by two main organic acids, malic and tartaric acids (later called malate and tartrate in this report). Malate is involved in malolactic fermentation, a necessary step for red wine which gives flexibility and roundness. Tartrate is the major wine acid that determines wine freshness and aging potential (Conde *et al.*, 2007).

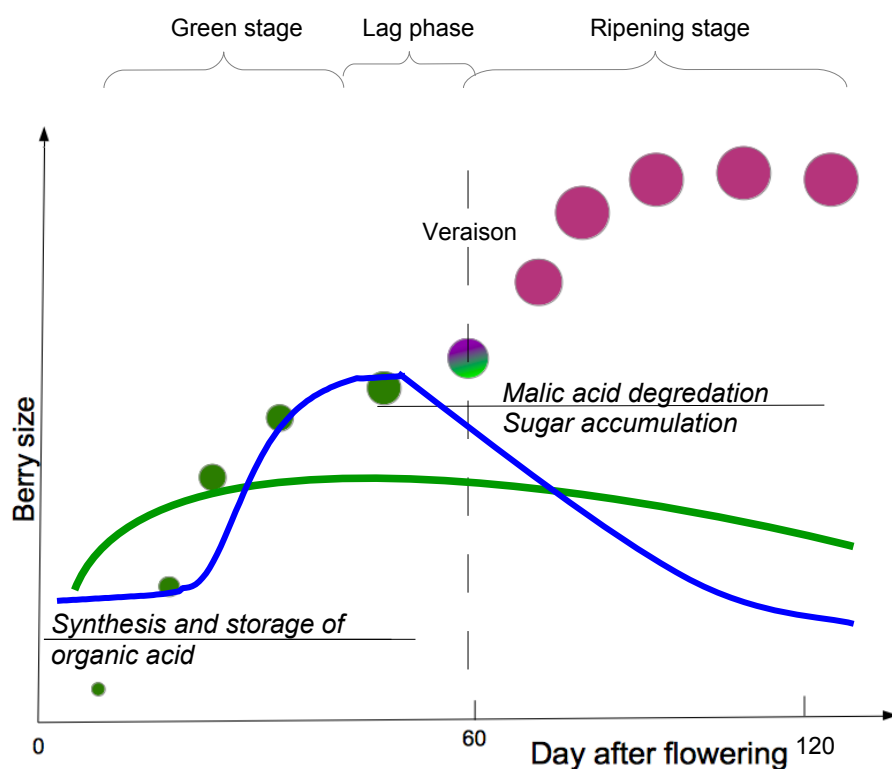


Figure1 Grapevine berry's development with theoretical periods of organic acids metabolism. Inspired from Coombe et al. (2001). Green and blue curve present the relative trends of tartrate and malate content evolution, respectively, in grape berry.

Grape berries show a double sigmoid curve of development (Figure 1), with two growth phases separated by a lag phase. The transition from the lag phase to the second growth phase is termed “veraison”, during which berries start to change colour from green to red or light yellow, depending on the varieties. Throughout the pre-veraison period, named herbaceous (or green) stage, berries accumulate high levels of malate and tartrate (Kliewer 1965; Saito and Kasai 1968). Tartrate exhibits an accumulation between the 4th and 6th week post-flowering (DeBolt *et al.*, 2006) and then there is no evidence of further production or degradation in ripening berries. Conversely, malate shows a pre-veraison accumulation phase and a continuous decrease post-veraison (Kliewer 1965; Sweetman *et al.*, 2009). Post-veraison decrease of malate content is attributed to: (i) an increase of respiratory rate, malate being the main respiratory substrate at this stage, (ii) dilution effect due to the increase of berry volume (iii) vacuolar K⁺ increase during ripening (Kliewer 1965).

1.2 Malate and tartrate metabolism in grape berry development

Despite their close molecular structure (weak di-acids), tartrate and malate are involved in different metabolic pathways during berry development, but their vacuolar storage is based on a common mechanism. Tartrate is a metabolic dead end and its degradation is not known to occur (Saito and Kasai 1968). Its synthesis is the result of L-ascorbic acid catabolism through the Smirnoff-Wheeler pathway (Melino *et al.*, 2009). The key enzyme of its synthesis, L-idonate dehydrogenase, that catalyses the conversion of L-idonate to 5-keto D-gluconic acid is present early in green stage, concomitantly with the tartrate synthesis peak (Debolt *et al.*, 2006, Deluc *et al.*, 2007). Malate content rises progressively and peaks at the beginning of the lag phase. Its major biosynthetic pathway involves the irreversible carboxylation of phosphoenolpyruvate (PEP) into oxaloacetate (OAA) by phosphoenolpyruvate carboxylase (PEPC). OAA is then reduced to malate by cytosolic NAD-dependent malate dehydrogenase (NAD cytMDH, Hawker, 1969; Sweetman *et al.*, 2009; Martinez-Esteso *et al.*, 2013).

Once produced, tartrate and malate are stored in the vacuole of the berry mesocarp cells. The mechanism that allows the accumulation of tartrate and malate in the vacuole has been described as an ‘acid-trap’ (Sweetman *et al.*, 2009, Fontes *et al.*, 2011, Etienne *et al.*, 2013). De Angeli and coworkers (2013) identified a tonoplastic channel, VvALMT9 that can let dianionic forms of malate and tartrate cross tonoplast, according to their cytosol-vacuole concentration gradient. VvALMT9 expression levels increase during berry herbaceous phase, concomitantly to the import of tartrate and malate in the vacuole (De Angeli *et al.*, 2013). Fruit vacuoles are highly acidic (pH of 2.5), during green stage, due to tonoplast bound proton pump activities (V-ATPase and V-PPase, Terrier

et al., 2001; Pilati *et al.*, 2007; Martinez-Esteso *et al.*, 2013). This makes tartrate and malate nearly fully protonated inside vacuoles because of their low pKa values (Malate; pKa1 = 3.40, pKa2 = 5.11, Tartrate; pKa1 = 2.98, pKa2 = 4.34, De Angeli *et al.*, 2013, Etienne *et al.*, 2013) and thus impermeant across tonoplast, effectively trapping them in the vacuole.

During berry ripening, malate exiting the vacuole can fuel the TCA cycle, thanks to dicarboxylate-tricarboxylate mitochondrial carriers (VvDTC1, VvDTC2, VvDTC3, Regalado *et al.*, 2013). Then it can be oxidized into OAA or pyruvate, respectively by NAD-dependent malate dehydrogenase (NAD-mtMDH) and NAD-dependent malic enzyme (NAD-mtME) (Sweetman *et al.*, 2009; Sweetlove *et al.*, 2010; Etienne *et al.*, 2013; Famiani *et al.*, 2014; Walker *et al.*, 2015). Besides massive sugar import from phloem unloading (Fontes *et al.*, 2011; Dai *et al.*, 2009), hexose accumulation in berry mesocarp cells may also occur through gluconeogenesis from malate. The requirement of PEP for gluconeogenesis may in theory be fulfilled by two enzyme couples in cytosol: the NAD cMDH/PEPCK (PEP Carboxykinase) and the NADP cME (NADP-dependent malic enzyme)/PPDK (pyruvate orthophosphate dikinase) that both need malate as substrate. However, only the former has been shown to be active in ripening berries, PPDK activity being totally absent (Famiani *et al.*, 2005; 2014; 2016; Terrier *et al.*, 2005; Pilati *et al.*, 2007; Sweetman *et al.*, 2012; Martinez-Esteso *et al.*, 2013; Walker *et al.*, 2015).

Temperature exerts its effects on berry's organic acid content mainly by impacting malate metabolism. An increase of malate content before veraison under warming may be explained by a stimulation of both synthesis and storage since vacuolar pump activities and *VvALMT9* expression are increased by a rise of temperature (Terrier *et al.*, 1998; Etienne *et al.*, 2013; Sweetman *et al.*, 2014; Rienth *et al.*, 2016). Once malate content reaches its higher value, just before veraison and throughout ripening, high temperature increases the degradation rate of malate (Sweetman *et al.*, 2014). In mitochondria, activity and transcript level of NAD-ME increase with heat treatments, particularly at veraison and during ripening (Ruffner *et al.*, 1976; Sweetman *et al.*, 2014).

1.3 Modelling organic acids evolution during berry development and ripening

Even if the global trend of the temperature impact on organic acids in grape is known (i.e. a lower content of malate at ripeness in warmer conditions), viticulture is still lacking tools that could predict with reasonable accuracy the evolution of organic acid content in response to temperature. Process-based models have the potential to produce such tools. The SUGAR model, where sugar amount evolution is based on carbon import, partitioning and depletion, is all ready applied on

peach and grape berry and predict accurately evolution of sugar content in fruit development (Génard and Souty, 1996; Dai *et al.*, 2009). Process-based model describing organic acids amount have been developed for peach and banana fruit. The citrate accumulation model for banana fruit, proposed by Etienne *et al.* (2015) is based on its synthesis through a mechanistic model of TCA cycle of Lobit *et al.* (2003). This can be achieved because citrate vacuolar accumulation is likely to be limited by its synthesis, its vacuolar uptake needing secondary active transport (Etienne *et al.*, 2013). In comparison, vacuolar malate uptake is based on facilitated diffusion and acid-trap (De Angeli *et al.*, 2013; Etienne *et al.*, 2013). Therefore the malate model predicting malate accumulation in peach and in banana fruits is based on its vacuolar storage (Lobit *et al.*, 2006; Etienne *et al.*, 2014). Lobit *et al.* (2002) computed vacuolar pH through its acid/base composition, that is solved by an iterative resolution process, allowing malate to be predicted over fruit development by thermodynamic constraints of its vacuolar transport (Lobit *et al.*, 2006; Etienne *et al.*, 2014). An important outcome of this pre-existing model is that it accurately predicts the negative effect of temperature on malate content in both peach and banana fruits. The thermodynamic model of malate transport of Lobit *et al.* (2006) and Etienne *et al.* (2014) might be adapted to predict grape berry's organic acid accumulation. Despite their different metabolism, we hypothesized that tartrate and malate have in common the process of their vacuolar storage. Thus a global thermodynamic approach of their vacuolar transport has the potential to describe malate and tartrate content accumulation and their response to temperature during berry development.

Therefore, the aim of this work is to develop, calibrate and validate a process-based model for predicting malate and tartrate content variations in response to temperature, during grapevine berry development and maturation, based on the pre-existing model of Lobit *et al.* (2006)

II Material & Method

2.1 System description and hypothesis

The model will assume grape berry as one single large cell, where an acidic vacuole allows the acid trap mechanism, implying that malate and tartrate cross tonoplast by facilitated diffusion at any developmental stage. In order to simplify the system, only vacuolar H⁺-ATPase is considered to be involved in protons pumping into the vacuole, V-PPase contribution is considered negligible. The model is based on 3 concepts: (i) thermodynamic equilibrium across tonoplast of di-anionic forms of organic acids established by vacuolar H⁺-ATPase, (ii) acidic properties of organic acids

that permit the vacuolar acid-trap, (iii) vacuolar pH computation through its ionic composition, taken from Lobit et al. (2006), and Etienne et al. (2014). The key assumption is thermodynamic equilibrium, which implies steady-state: the free energy associated with the coupled ATP hydrolysis/proton transport reaction is zero at every given time. Therefore, the model will compute pH and/or organic acid content at every sampling stage. Because of the relative importance of the vacuole volume in the cell, protons and organic acids content of cytosol are considered negligible, compared to vacuolar ones. Thus, vacuolar acidity is assumed to be the global measured acidity, as well as global organic acids content measured (mg g⁻¹) divided by their respective molar masses (tartrate: 150 g mol⁻¹, malate: 134 g mol⁻¹) are assumed to be vacuolar organic acids concentrations.

2.2 Mathematical development of the model

The tonoplast energisation by H⁺-ATPase activity of proton pumping is derived from Nernst equation, which can describe the variation of free energy of vacuolar H⁺-ATPase that results from a coupling between ATP hydrolysis and proton transport (Lobit *et al.*, 2006, Etienne *et al.*, 2013; 2014):

$$\Delta G_{ATPase} = \Delta G_{ATP} + nF\Delta\Psi - nRT \ln(10) \Delta pH \quad (1)$$

where R is the perfect gas constant (8.314 J mol⁻¹K⁻¹), F the Faraday constant (9.65 10⁴ C mol⁻¹), T is the temperature (K), ΔpH is the differences in pH between vacuolar (pH_{vac}) and cytosol (pH_{cyt}), ΔG_{ATP} is the reference free energy of ATP hydrolysis, the electric potential gradient (electromotive force) across tonoplast ($\Delta\Psi = \Psi_{vac} - \Psi_{cyt}$), and n is the stoichiometry of pumping reaction :

$$n = n_0 + \alpha(pH_{vac} - 7) + \beta 10(pH_{cyt} - 7) \quad (2)$$

where n_0 (4), α (0.3) and β (-0.12), all dimensionless, came from Lobit et al. (2006). At thermodynamic equilibrium, $\Delta G_{ATPase} = 0$ and $\Delta\Psi$ can be expressed as:

$$\Delta\Psi = (nRT \ln(10) \Delta pH - \Delta G_{ATP})/nF \quad (3)$$

Because the transport of malate across the tonoplast is passive, free energy variation of di anionic malate (Mal²⁻) across tonoplast depends on the electric potential gradient and on ratio of activities of Mal²⁻ in the two compartments:

$$\Delta G_{Mal^{2-}} = -2F\Delta\Psi + RT \ln \left(\frac{(Mal^{2-})_{vac}}{(Mal^{2-})_{cyt}} \right) \quad (4)$$

similarly, the Gibbs free energy variation of tartrate across tonoplast is:

$$\Delta G Ta^{2-} = -2F\Delta\Psi + RT \ln \left(\frac{(Ta^{2-})_{vac}}{(Ta^{2-})_{cyt}} \right) \quad (5)$$

The organic acid activity (Mal^{2-}) in a solution is $(Mal^{2-}) = \alpha(Mal^{2-}) [Mal^{2-}]$

where $\alpha(Mal^{2-})$ is the activity coefficient which is equal to 1 in diluted solution (cytosol for instance) or is a function of ionic strength I (derived from Güntelberg relation, Etienne *et al.*, 2013, Lobit *et al.*, 2006, Etienne *et al.*, 2014) in concentrated solution (vacuole):

$$\alpha(Mal^{2-}) = \alpha(Ta^{2-}) = 10^{-0.509 z^2 \frac{\sqrt{I}}{(1+\sqrt{I})} - 0.31} \quad (6)$$

where z is the corresponding ion valence. Tartrate and malate are almost fully protonated inside the vacuole ($2.5 \leq pH_{vac} \leq 3.5$), thus relations between vacuolar di-anionic form concentrations and total concentration are linked to dissociation relations. An acid and its base are involved with their dissociation constant and with pH in Henderson–Hasselbalch relation: $K_i = [H^+] [A^-]/[AH]$. This allows to compute the relation between the concentration of a given molecular form (here the di anion form) and the total concentration of malate:

$$[Mal]_{vac\ total} : [Mal^{2-}]_{vac} + [MalH^-]_{vac} + [MalH_2]_{vac}$$

So, for malic and tartaric acids:

$$[Mal^{2-}]_{vac} = [Mal]_{vac\ total} M(h) \text{ with } M(h) = (K_{m1} K_{m2}) / (K_{m1} K_{m2} + h K_{m1} + h^2) \quad (7a)$$

$$[Ta^{2-}]_{vac} = [Ta]_{vac\ total} T(h) \text{ with } T(h) = (K_{t1} K_{t2}) / (K_{t1} K_{t2} + h K_{t1} + h^2) \quad (7b)$$

K_{m1} , K_{m2} , K_{t1} and K_{t2} are acidic apparent constant of malate and tartrate, respectively; computed through activity coefficients of the corresponding acids and bases: $K_{mi} = (\alpha(AH)/\alpha(A^-)) * K_{ai}$

K_{ai} is the constant of the acid i ($K_{ai} = 10^{-pK_{ai}}$) and $h = 10^{-pH} = [H^+]$

For malate $pK_{a1} = 3.40$, $pK_{a2} = 5.11$; for tartrate $pK_{a1} = 2.98$, $pK_{a2} = 4.34$. Computations are made with the assumption of the thermodynamic equilibrium at steady state, $\Delta G Mal^{2-}$, $\Delta G Ta^{2-}$ and $\Delta G ATPase$ are equal to zero. Thus:

$$[Mal]_{vac\ total} = [Mal^{2-}]_{cyt} \left(\frac{1}{M(h)} \right) e^{(2F\Delta\Psi/RT)} \frac{1}{\alpha(Mal^{2-})_{vac}} \quad (8a)$$

$$[Ta]_{vac\ total} = [Ta^{2-}]_{cyt} \left(\frac{1}{T(h)} \right) e^{(2F\Delta\Psi/RT)} \frac{1}{\alpha(Ta^{2-})_{vac}} \quad (8b)$$

These equations describe vacuolar content of organic acids as a relation of three gradients between vacuole and cytosol: the pH gradient, organic acids concentration gradients and the electromotive force ($\Delta\Psi$). The vacuolar pH is involved in $\Delta\Psi$ generation (equation (2)) and in organic acid dissociations according to M(h) and T(h)(equation (8a) & (8b)). Fruit vacuolar pH, leading to global fruit total acidity, has been predicted by the pH model of Lobit et al. (2002). It allows to predict pH and ionic strength I with vacuolar ionic composition as inputs, through two main equations: (i) electro-neutrality and (ii) ionic strength computation :

$$(i) \sum z_i [\text{anion}] = \sum z_i [\text{cation}] \quad (9a)$$

$$(ii) I = \frac{1}{2} \sum z_i^2 [\text{ioni}] \quad (9b)$$

Cations, K^+ and H^+ , and anions, Mal^{2-} , $MalH^-$, Ta^{2-} , TaH^- and HO^- , are used in the model and their dissociation equations (relation with pH based on acidic properties, see above) are linked to vacuolar pH.

2.3 Model inputs and data sets

Input variables required to develop the models are temperature (T; °K), pH, malate (mol L⁻¹), tartrate (mol L⁻¹) and potassium (mol L⁻¹). Four data sets comprising five genotypes of *V. vinifera* have been used for model development and extracted from literature using PlotDigitizer (<http://plotdigitizer.sourceforge.net/>) (Table 1). A first data set from Kliewer et al (1965) with Cardinal and Pinot Noir genotypes and two daily temperature regimes (with daily temperature of 20 °C and 30 °C, night temperature of 15 °C), at five sampling stages. A second data set from Soyer (Soyer *et al.*, 1993) uses Cabernet Sauvignon vines submitted to two potassium fertilizations “K60” and “K120” with control K0, corresponding to an annual application of 0, 60 and 120 kg K₂Oha⁻¹. A third data set, using Sangiovese was extracted from Bobeica et al. (2015) and consist in two leaf-to-fruit ratios, implemented by keeping 3 or 12 leaves per shoot (designated “3L” and “12L”). The last data set is the one acquired in Etchebane et al. (2010) with Grenache cultivar treated by different

Table 1 : Genotypes from four data sets used to develop the model.

Genotype	Treatment	Reference
Grenache	Irrigation, source to sink ratio	Etchebane <i>et al.</i> , 2010
Cabernet Sauvignon	Potassium fertilization	Soyer <i>et al.</i> , 1993
Sangiovese	source to sink ratio	Bobeica <i>et al.</i> , 2015
Pinot Noir	Temperature	Kliewer <i>et al.</i> , 1965
Cardinal	Temperature	Kliewer <i>et al.</i> , 1965

irrigation levels and various modification of leaf-to-fruit ratio. Four of the six treatments were used, depending on potassium measurement availability.

2.4 Work flow and model construction strategy

Model construction was made in 3 different steps: the first step was the implementation of a malate model which compute vacuolar malate with measured pH, and temperature. Then, a tartrate model was created and coupled to the malate model to generate a global model able to compute both organic acid concentrations in a single step. In the second step, the pH model (Lobit *et al.*, 2002) had been adapted to grape in order to compute vacuolar pH, with inputs of tartrate, malate, T and K⁺. Then, in a third (and final) step, the models for malate, tartrate and pH were integrated in order to predict both vacuolar organic acid contents and vacuolar pH using only measured T and K⁺ content as inputs (Figure 2).

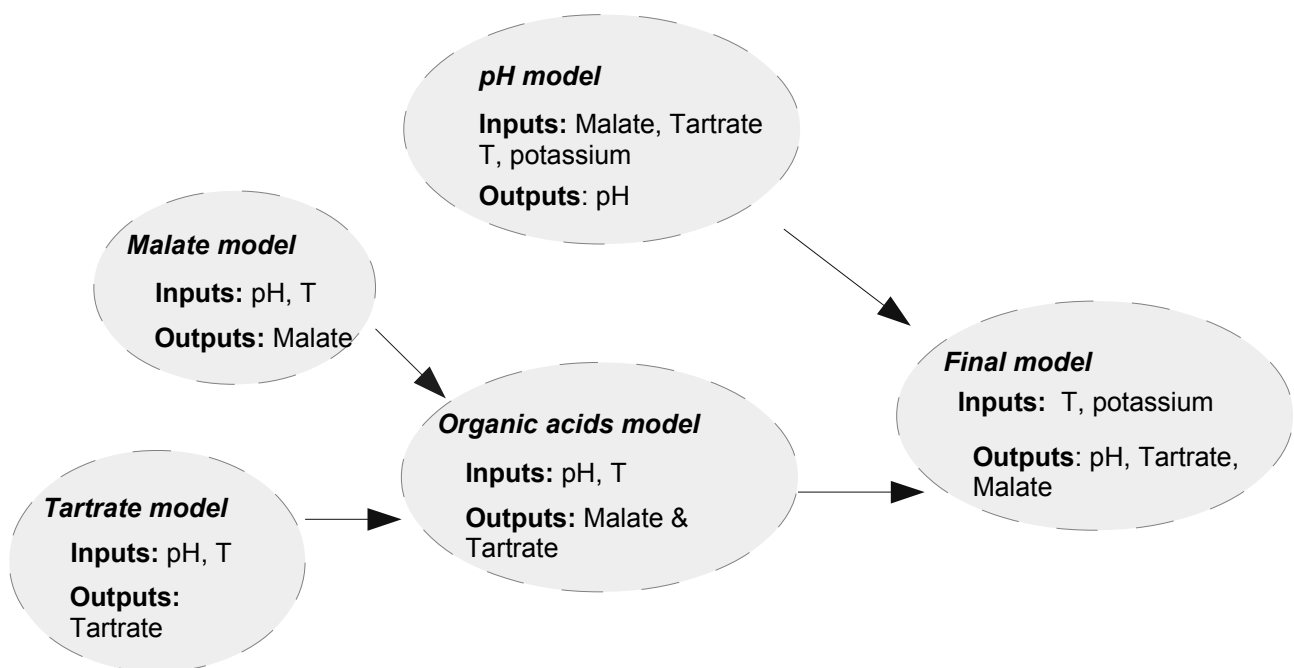


Figure 2. Strategy to develop the model. Step 1: Malate predicting model, then implementation of a tartrate predicting model and coupling with the malate model. Step 2: Adaptation the pH model of Lobit et al. (2002) to grape berry and step 3: coupling to obtain the final model. T is temperature.

2.5 Model solving

Genotype-dependant parameters were estimated by fitting computed values to observed ones using “nls” function or “optimx” function of R software. In addition, “nleqslv” function (package “nleqslv”) was used to solve a system of non linear equations when combining the acid (malic and tartaric) models with the pH model.

The goodness-of-fit was computed by calculating RRMSE (Relative Root Mean Squared Error): $RRMSE = RMSE/m$, with RMSE (Root Mean Squared Error, using R package “Metrics”) $RMSE = \sqrt{(\sum (x_{ij} - y_{ij})^2/nb)}$, where x_{ij} is a specific measured variable and y_{ij} is the correspondent output from any model. m is the mean of observed variable. nb is the number of data of x_{ij} . The smaller the RRMSE, the better the accuracy of the fit. All data analyses were performed with R software (R Development Core Team, <http://www.R-project.org>). Figures are made using R package “ggplot2”.

III RESULTS

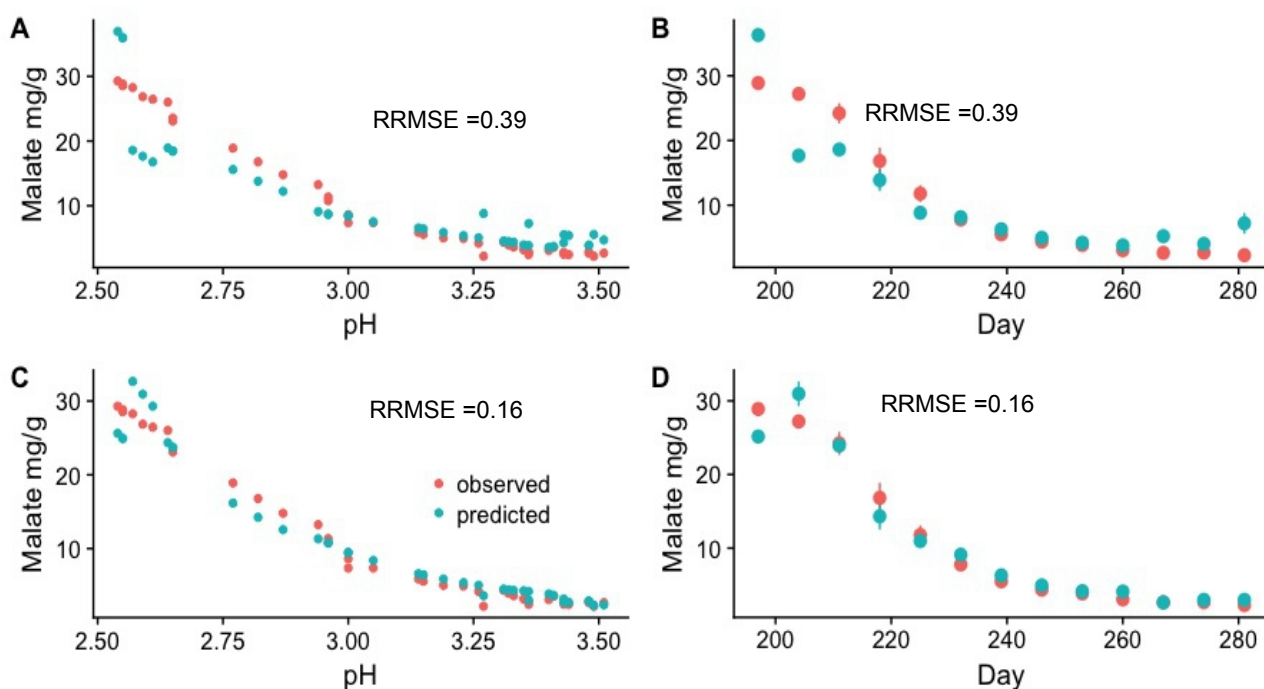


Figure 3 Malate prediction, computed via measured pH and temperature. Malate model applied on Cabernet Sauvignon data set, on treatment K60. Red dots are observed value, blue dots are model simulation. Comparison of fits between A, B where $\Delta GATP$ is constant ($-57.8 \text{ kJ mol}^{-1}$) and C, D where $\Delta GATP$ is linear through the CIX function ($a = -298 \text{ J mol}^{-1} \text{ K}^{-1}$, $b = 29815 \text{ J mol}^{-1}$), both with $[\text{Mal}]_{\text{cyt}} = 0.65 \text{ mM}$. Evolution of malate concentration over pH (A, C) where dots are each sampling value and correspondent predicted value, and over days of the year (B, D) where values are mean $\pm$ sd ($n=3$). RRMSEs are indicated in each graph.

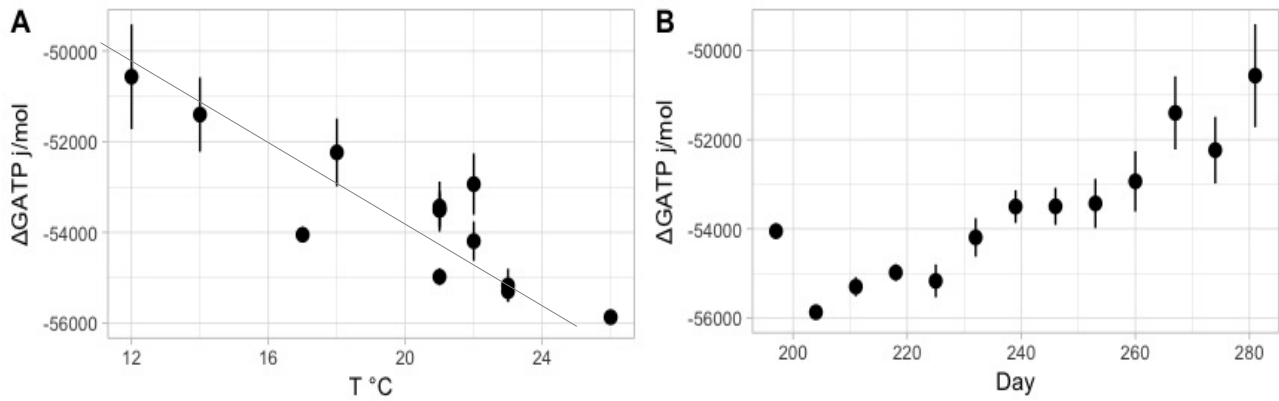


Figure 4 $\Delta GATP$ evolution over temperature (A) and days (B) by the inversion of Malate model, applied on Cabernet Sauvignon data set for all treatments. Computation of $\Delta GATP$ through observed Malate and pH with $[Mal]_{cyt} = 1$ mM. Data are mean $\pm$ sd ($n=13$). Linear regression on A gives with CIX relation : $a = -349 \text{ J mol}^{-1} \text{ K}^{-1}$, $b = 48817 \text{ J mol}^{-1}$, $R^2=0.649$

3.1 Calibration of the malate predicting model

The malate predicting model corresponds to the equations (3), (7a) and (8a), written in R. In the model, vacuolar malate concentrations is calculated with measured pH (considered as vacuolar pH) and temperature as inputs and $\Delta GATP$ and $[Mal]_{cyt}$ as unknown parameters. After parameterization, we found an underestimation of vacuolar malate at low pH and an overestimation at higher pH (Figure 3A and 3B). In order to improve the model behaviour, a model inversion was used to plot $\Delta GATP$ evolution as a function of berry developmental stage or of temperature, for a constant $[Mal]_{cyt}$ (Figure 4). To do so, for each sampling date, $\Delta GATP$ is computed using measured malate, pH, and temperature throughout inversion of equation (2) and (3). A linear relation can be observed between $\Delta GATP$ and temperature (Figure 4A), which was named the “CIX” (arbitrary designation) relation: $\Delta GATP = a \cdot T + b$ with a ($\text{J mol}^{-1} \text{ K}^{-1}$) and b (J mol^{-1}). When introducing the CIX function back to the model, it significantly improved the fitness accuracy (figure 3 C, 3D) for describing the accumulation of malate all over the berry development.

3.2 Tartrate predicting model

Tartrate can be predicted in a similar manner than malate as the malate predicting model: equations (3), (7b) and (8b) written in R allowed to compute tartrate evolution with the input of measured pH. The simulated results showed large discrepancy with observed values when a constant $\Delta GATP$ and $[Ta]_{cyt}$ was used in the model (Figure 5A and D). The goodness-of-fit was slightly improved when including the variation of $\Delta GATP$ as a function of temperature as described

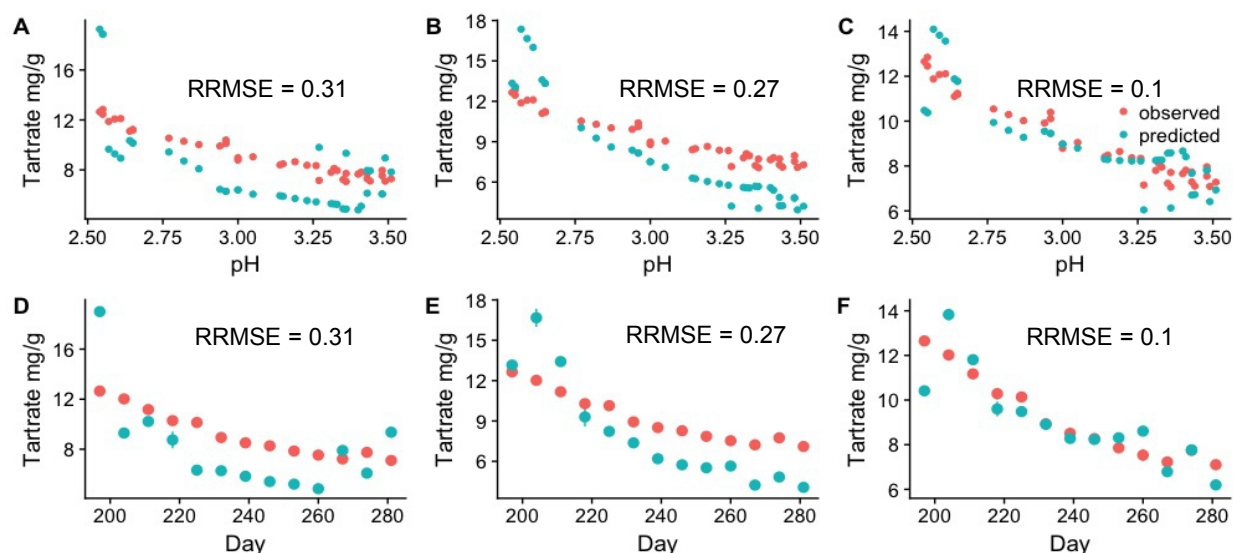


Figure 5 Tartrate prediction, computed via measured pH and temperature. Calibration progression of Tartrate model applied on Cabernet Sauvignon data set, on treatment K60. Red dots are observed value, blue dots are model simulations. Comparison of fits between (A,D) $\Delta GATP$ is constant ($-57.7 \text{ kJ mol}^{-1}$), (B,E) $\Delta GATP$ varies linearly with temperature through CIX function ($a = -298 \text{ J mol}^{-1} \text{ K}^{-1}$, $b = 29815 \text{ J mol}^{-1}$), both (A,D and B,E) have $[Ta]_{\text{cyt}} = 3.6 \text{ mM} = 0.54 \text{ mg g}^{-1}$, and (C,F) $\Delta GATP$ varies linearly with temperature, through CIX function (same a and b than B,E) but cytosolic tartrate is linear ($T1 = 0.0032 \text{ M}$, $T2 = -0.053 \text{ M}$). Graph with tartrate concentration over pH (first row) have values corresponding to each sampling value and correspondent predicted value. Graph with tartrate concentration over day (second row, day of the year) have values corresponding to mean $\pm$ sd ($n=3$). RRMSE are indicated in each graph.

with the CIX function (Figure 5B, E). Further improvement was obtained by model inversion, which allowed to compute $[Ta]_{\text{cyt}}$ through observed pH and tartrate, (inversion of equation 7b, supp Figure 1). A linear relationship was observed for a progressive increase of $[Ta]_{\text{cyt}}$ as a function of pH_{vac} , which could be formulated with $T1$ and $T2$ as $[Ta]_{\text{cyt}} = T1 * pH_{\text{vac}} + T2$ (Figure 8: cytosolic evolution of organic acids). When integrating both CIX and the variation of $[Ta]_{\text{cyt}}$ into the model, tartrate simulation showed a much better RRMSE (figure 5 C, F), indicating a much accurate prediction of tartrate content evolution.

3.3 Coupling organic acid prediction and temperature response

The above section showed that the malate and tartrate models predicted accurately the two organic acids, when working independently. We further tested whether a coupled model combining both acid models can predict the acid composition in berry under various experimental conditions. To this end, the dataset from Kliewer et al. (1965) with two genotypes and two temperature regimes

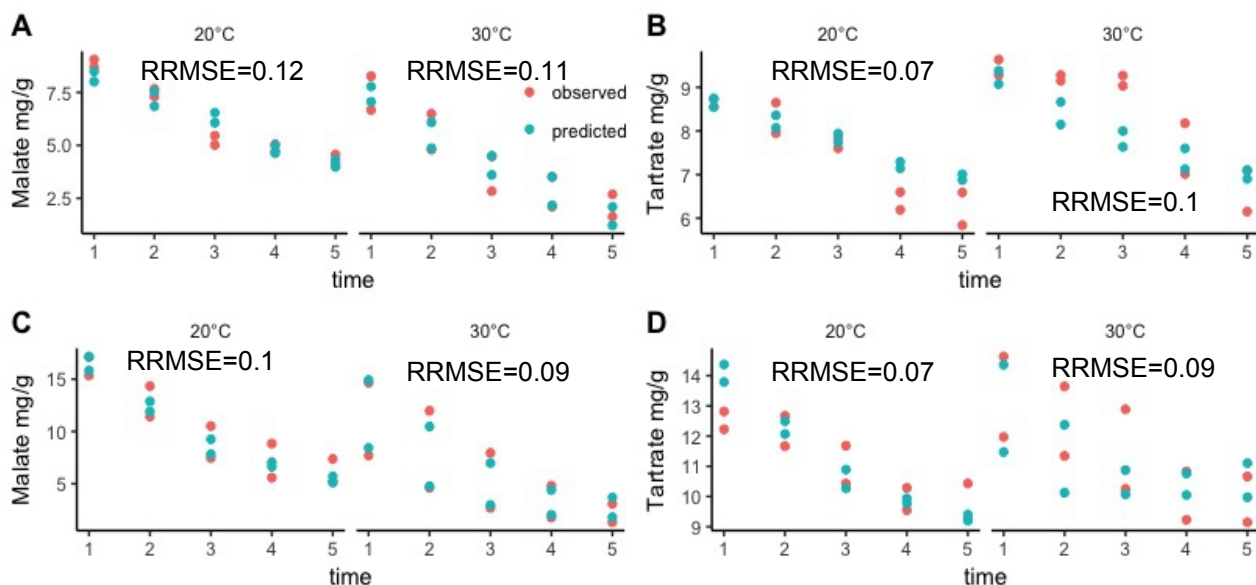


Figure 6. Prediction of tartrate and malate and their temperature response applied on Cardinal (A,B) and Pinot Noir (C,D). Organic acids computation are made using a, b, M1, M2, T1, T2 and cte for calibration. RRMSE are indicated in each graph. Red dots observed; blue dots predicted. Time is week after temperature treatment. For a night temperature at 15°C, vines were either submitted to 20°C or 30°C.

were used. The coupled model correctly predicted the negative effects of high temperature on malate concentration in cvs Cardinal and Pinot Noir (Figure 6A and 6C). Moreover, it also reproduced the absence of responses of tartrate to high temperature in both cultivars (Figure 6B and 6D).

Model analysis was used to explore the potential regulating points for the differential responses of malate and tartrate to high temperature. This was done by assessing the relative importance of each

Table 2 : Example of calibration evolution and consequences on prediction of the temperature response observed in Pinot Noir dataset. Grey box are RRMSE (goodness of fit) and white box are RRMSEP (quality of prediction).

	Parameters					
	a	b	M1	M2	T1	T2
PN 20&30	0	-56000	0	9.56 10 ⁻⁴	0	1.055 10 ⁻²
PN 20&30	-229	11628	0	9.56 10 ⁻⁴	0	1.055 10 ⁻²
PN 20&30	-229	11628	-7.57 10 ⁻⁴	3.2 10 ⁻³	2.11 10 ⁻³	4.3 10 ⁻³
PN 30	-229	11628	-8.45 10 ⁻⁴	3.49 10 ⁻³	-3.87 10 ⁻⁵	1.16 10 ⁻²
PN 20	-229	11628	-5.14 10 ⁻⁴	2.5 10 ⁻³	5.67 10 ⁻³	-6.34 10 ⁻³

	RRMSE & RRMSEP					
	Tartrate			Malate		
	20&30	20	30	20&30	20	30
PN 20&30	0.39	0.24	0.49	0.32	0.16	0.56
PN 20&30	0.11	0.1	0.11	0.23	0.16	0.35
PN 20&30	0.09	0.07	0.1	0.12	0.12	0.11
PN 30	0.12	0.15	0.08	0.12	0.12	0.10
PN 20	0.13	0.04	0.17	0.14	0.11	0.19

parameter to the goodness-of-fit of the model to the observed results. When varying a parameter can significantly improve the goodness-of-fit of an output (e.g. malate), it indicates that the parameter plays an important regulating role in the output variable. The goodness-of-fit was quantified by RRMSE and RRMSEP (cross validation, Table 2B) by modifying the three parameters, including $\Delta GATP$, $[Mal]_{cyt}$, $[Ta]_{cyt}$. Results showed that the variation of $\Delta GATP$ as a function of temperature was of most important for an accurate simulation of tartrate in response to temperature. Including a varying $\Delta GATP$ increased the goodness-of-fit from a RRMSE at 0.39 to 0.11. The variation of cytosol Malate and tartrate as a function of vacuolar pH is essential to have an accurate simulation of malate in response to temperature (Table 2B), with RRMSE changing from 0.23 to 0.12.

3.4 pH predicting model

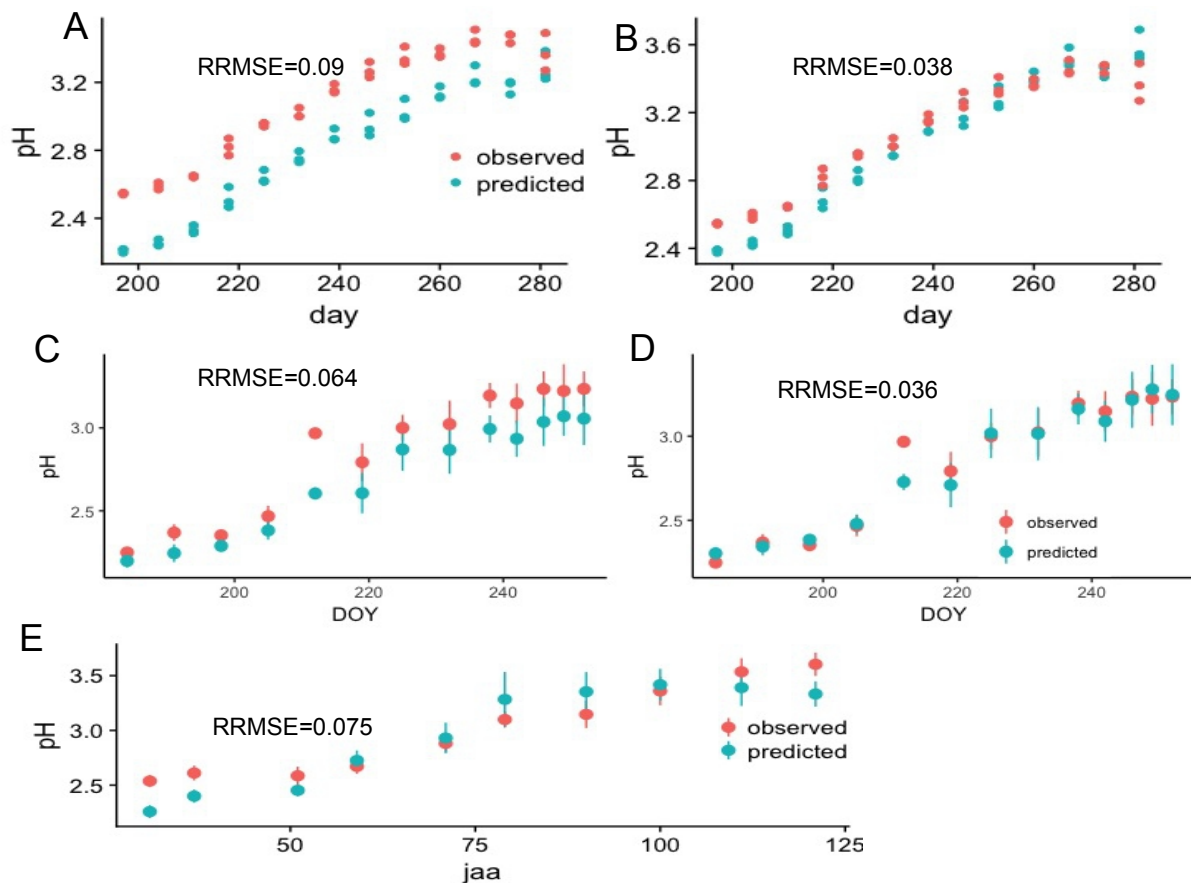


Figure 7 Prediction of vacuolar pH using pH model of Lobit et al. (2002). A: pH computation for Cabernet Sauvignon using Lobit's calibration for peach. B: pH computation for Cabernet Sauvignon using parameter obtain by model calibration specific for grape ('cte' = 8 mM). C: pH computation for Sangiovese using Lobit's calibration for peach. B: pH computation for Sangiovese using parameter obtain by model calibration specific for grape ('cte' = 4.72 mM). E: pH computation for Grenache using lobit's parameter, no grape-specific parameter could be fitted. Red dots observed; blue dots predicted.

The above organic acid models used fruit pH as an input; however, grape growers are also interested in the prediction of pH itself. The pH model developed in peach and banana (Etienne *et al.*, 2014) was therefore adapted for grape berries with modifications taking into account grape specificities. The main difference in the present grape model is the fact that vacuolar ion composition is constituted by malate, tartrate, potassium and the water ionic products, while in peach and banana (Etienne *et al.*, 2014) quinate, citrate, phosphate, calcium and magnesium were main ion composition. After those modifications, the pH model correctly simulated the developmental profiles of pH in cv. Grenache (Figure 7E). However, the model simulation showed large discrepancy with observed values in Sangiovese (Figure 7C) and particularly in Cabernet Sauvignon with an important underestimation of 0.4 pH units (Figure 7A).

The failure of the pH model in Sangiovese and Cabernet Sauvignon might originate from the fact that calcium and magnesium are neglected in the ionic strength computation (equation (7b)). In order to have an idea about calcium and magnesium contribution to ionic strength in berry development, their relationships with K^+ were analysed (data not shown) from data set of Cabernet Sauvignon and Grenache where K^+ , Mg^{2+} and Ca^{2+} contents were measured over berry development. The sum of Mg^{2+} and Ca^{2+} remained relatively constant over fruit development, and therefore, a constant (new parameters called “cte”) that mimics the involvement of Ca^{2+} and Mg^{2+} in ionic strength computation, was added to the model. The addition of “cte” at 8 mM largely improved the prediction of pH for Cabernet Sauvignon (figure 7, A to B). The addition of “cte” at 4.72mM also improved the prediction of pH for Sangiovese (figure 7 C to D). However, addition of 'cte' value could not further improve the prediction of pH of Grenache.

3.5 Coupled model predicting malate, tartrate, and pH

After testing and calibrating the malate, tartrate, and pH models separately, we tested the prediction capacity of a coupled model that can predict malate, tartrate, and pH simultaneously. This is challenging because of the inter-dependency between pH and malate and tartrate. An iteration approach was applied to solve a set of non liner equations with three unknown variables (malate, tartrate, and pH). When the models of malate, tartrate, and pH were integrated, the simulations were far from observed values, most probably due to the inter-dependency between pH and the two organic acids. After various assessments, an addition of a liner variation of [Mal]cyt as a function of pHvac by M1 and M2 : $[Mal]_{cyt} = M1 * pH_{vac} + M2$ (Figure 9B) significantly improved the goodness-of-fit of the coupled model. The final model, with 7 parameters, was able

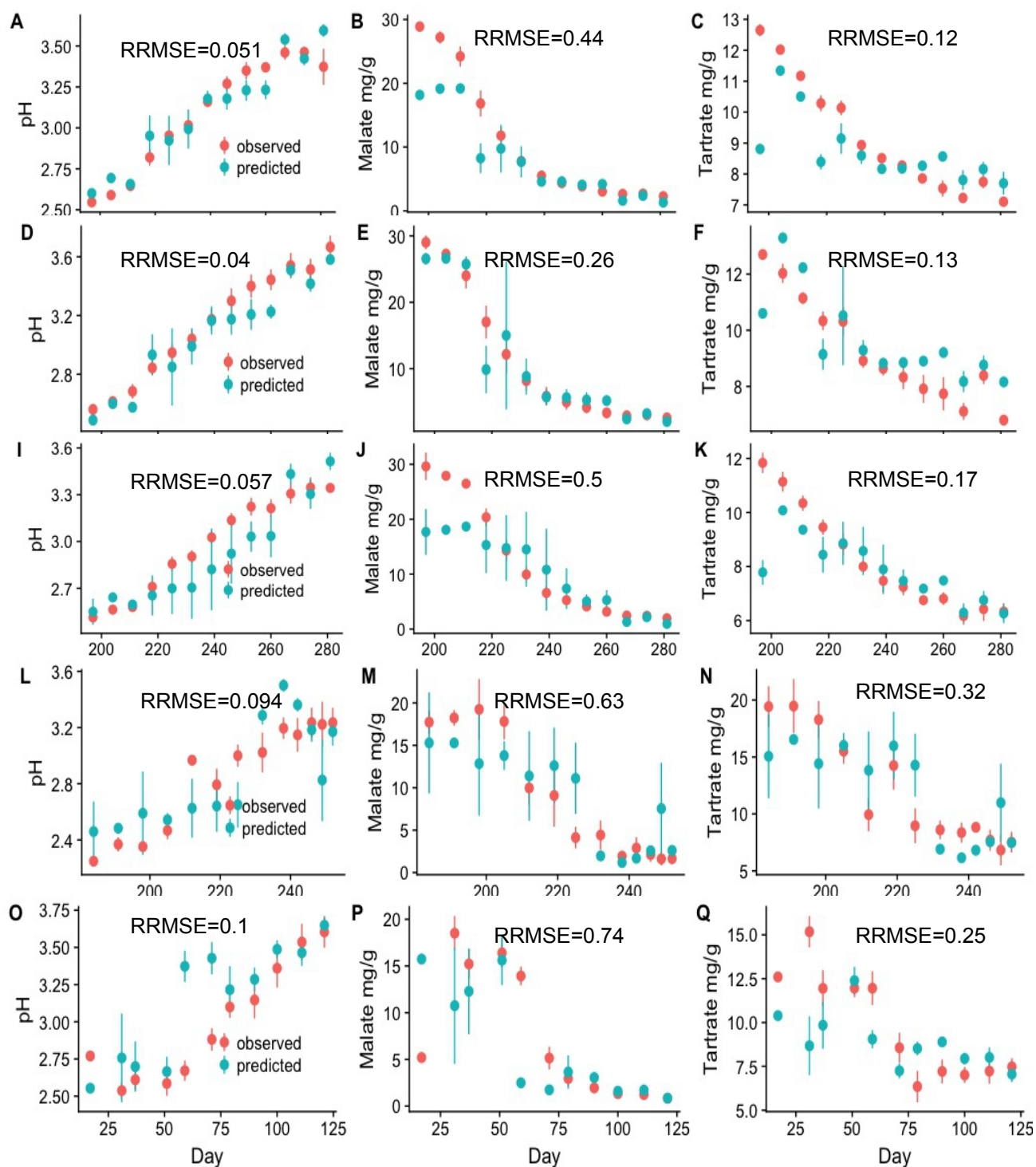


Figure 8 Final model applied on Cabernet Sauvignon treatment K60 (A,B,C), K120 (D,E,F), control KO (I,J,K), Sangiovese all treatments (L,M,N) and Grenache all treatments (O,P,Q). First column contains prediction of pH, second column contains prediction of malate and third column contains prediction of tartrate. Data are means $\pm$ sd ($3 \leq n \leq 8$). RRMSE are indicated in each graph. Red dots observed; blue dots predicted.

to give accurate predictions of pH, malate and tartrate for Cabernet Sauvignon (figure 8 A, B, C), Sangiovese (figure 8 D) and Grenache (figure 8 E).

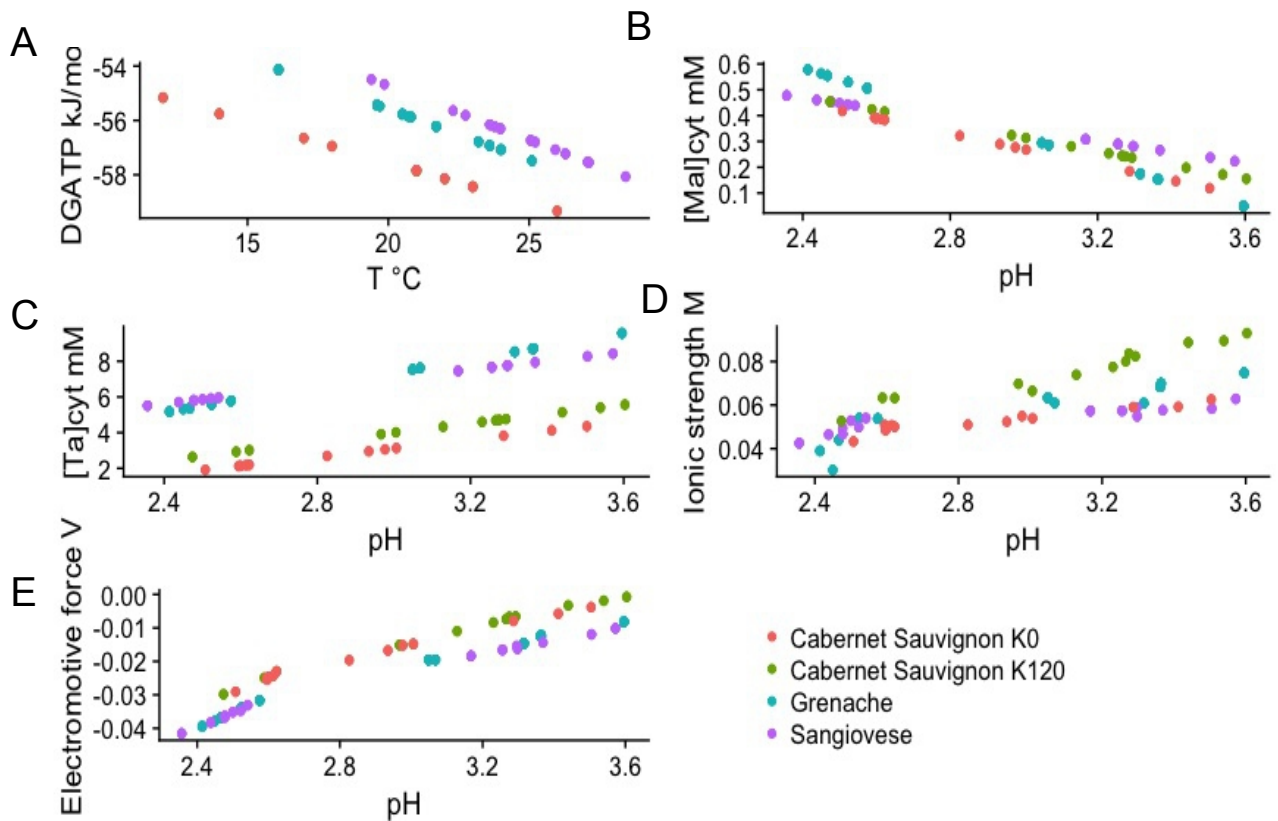


Figure 9 Overview of model's intermediate variables among data sets : final calibration results on Δ GATP, cytosolic concentration of malate and tartrate, and model's simulation of the electromotive force and ionic strength. Pinot Noir and Cardinal dataset are not represented here because they don't have potassium measurement which is necessary to run the final model.

With the validated model, we further explored the intermediate model variables to identify emergent properties of the model. The model is able to provide simulations of organic acid concentrations in the cytosol ([Mal]cyt, [Ta]cyt), the electro motive force $\Delta\Psi$, the ionic strength I and the Δ GATP, in function of fruit pH (Figure 9). Those variables are difficult to measure directly and the model simulations provide valuable estimations.

IV Discussion

4.1 Prediction of temperature response

Grape acidity is one of the main quality components for producing a balanced wine (Sweetman *et al.*, 2009). Therefore, it is appealing to have tools for predicting the levels of organic

acids in berries over development in response to changing climate conditions, so that the growers can decide the dates of harvest, particularly within the context of climate change (van Leeuwen *et al.*, 2016). Grape acidity is mainly determined by two organic acids, malate and tartrate (Sweetman *et al.*, 2009). Interestingly, the two acids showed different sensitivities to high temperature: malate was often reduced by high temperature, particularly during post-veraison, while tartrate was hardly affected by high temperature (Kliewer 1965). Our model reproduced accurately the differential responses of malate and tartrate to high temperature in two cultivars Cardinal and Pinot Noir (Figure 6). Model analysis indicated that the potential reasons underlying the differential sensitivity of malate and tartrate to high temperature are related with the relative concentration of malate and tartrate in the cytosol, [Mal]cyt and [Ta]cyt, respectively. In the model, [Mal]cyt varies from 0.1 to 0.6 mM, while [Ta]cyt varies from 2 to 8 mM (Figure 9). Therefore the gradient of tartrate across tonoplast is less pronounced than that of malate, which may in turn affect the efficiency of acid trap mechanism and resulting less sensitivity of tartrate to high temperature. However, the [Mal]cyt and [Ta]cyt are technically difficult to be quantified with direct measurements, therefore it remains uncertain whether the differences in cytosol concentrations reflect the real status in grape berry. Further experiments should be designed to make direct measurements of [Mal]cyt and [Ta]cyt in grape berry.

4.2 Is there a biological sense for the three new linear relations used in the model ?

Our models were developed based on previous models developed on peach and banana fruits (Lobit *et al.*, 2006; Etienne *et al.*, 2014). However, a direct application of the peach and banana acid model could not provide satisfactory simulations results. Therefore, several refinement of the model were conducted by including three linear relationships: 1) Δ GATP varies linearly as a function of temperature; 2) [Ta]cyt changes linearly with vacuolar pH; and 3) [Mal]cyt changes linearly with vacuolar pH. Although the Δ GATP was considered constant in the peach model (Lobit *et al.*, 2006), it changed between pre- and post-harvest for banana fruits (Etienne *et al.*, 2014). Here, a general linear relationship with temperature was observed in all treatments and cultivars studied in the current study, indicating the linear relationship between Δ GATP and temperature is robust. For this linear relationship, there are two parameters, a and b, with a for the slope and b for the intercept. The "a" seems to represent genotype-specific features, because it only differs between cultivars (Supp Figure 2); while "b" is controlled by [Mal]cyt and genotype.

Linear relationships of tartrate and malate cytosolic concentrations with vacuolar pH evolution allowed prediction of correspondent vacuolar organic acids content in the final step of model

development. It gave the evolution behaviour of the model between low and high pH (2,5 and 3,5) and improved cross-prediction. Strikingly, evolution of malate concentration in cytosol appeared to be conserved among genotypes, compared to the tartrate one. The increase in cytosolic tartrate concentration may rise the fact that the increased vacuolar pH diminishes acid trap on tartrate, leading to a progressive increase of tartrate in cytosol (Figure 9, [Tar]cyt). Direct measurements of [Mal]cyt and [Ta]cyt over berry development are needed to validate our model predictions.

4.3 Analysis of fruit acidness seems to highlight organic acid storage

The present model is based on storage of organic acids in the vacuole, since their accumulation may be limited by the thermodynamic constraints of their trans-tonoplast transport activities (Etienne *et al.*, 2013). Therefore the main mechanism of their accumulation may be the implementation of a proton gradient, and one can then imagine that organic acid storage is controlled by both vacuolar proton pumps and tonoplast channel to malate such as aluminium-activated malate transporter VvALMT9 (De Angeli *et al.*, 2013). This is in agreement with the fact that in apple fruits malate accumulation and vacuolar acidity are controlled by MdALMT9, MdALMTII and vacuolar proton pumps (vacuolar pyrophosphatase and vacuolar H⁺-ATPase) regulations (Hu *et al.*, 2017, Jia *et al.*, 2018). QTL (Quantitative Trait Loci) analysis linked to acidity in grape berry was not successful for tartrate content, but two QTLs controlling malic acid content and total acidity have been reported and found stable among two sampling years (Chen *et al.*, 2015). Even if the genes behind these QTLs have not been characterized yet (more than 134 found), some might be involved in transport. A QTLs analysis on a large F2 population highlight a malate transporter ALMT1 (Yang *et al.*, 2016) which may underline the importance of storage mechanism of malate content in grapevine berry. Finally, applying our model on whole grapevine F1 or F2 progenies with segregation for tartaric acid content and searching for stable QTLs for model parameters involved in tartrate accumulation may possibly prove helpful to gain insights on tartrate content genetic determinism. Such an approach has already been proposed for other complex trait in grape (Vivin *et al.*, 2017).

Conclusion

In conclusion, the models presented here predicted the accumulation of organic acids and pH among grapevine berry development. They were calibrated and validated in five

cultivars with treatments of high temperature, leaf-to-fruit modulation, as well as water stress treatments. The model predicted well the difference in acidity among different cultivars, in response to the various treatments. The different parameter used to calibrate the model in this work could be used in future genetic analysis to assist candidate gene identification for organic acid accumulation, and help the analysis of cultivar variability in acidity. This would pave the way to “model-assisted” genetic analysis and breeding strategies in grape.

Acknowledgements

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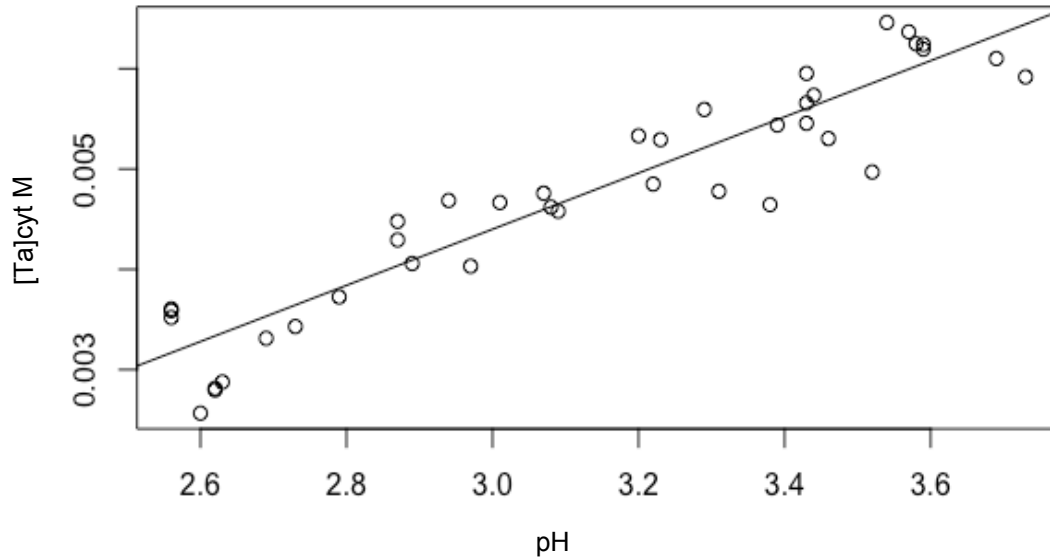
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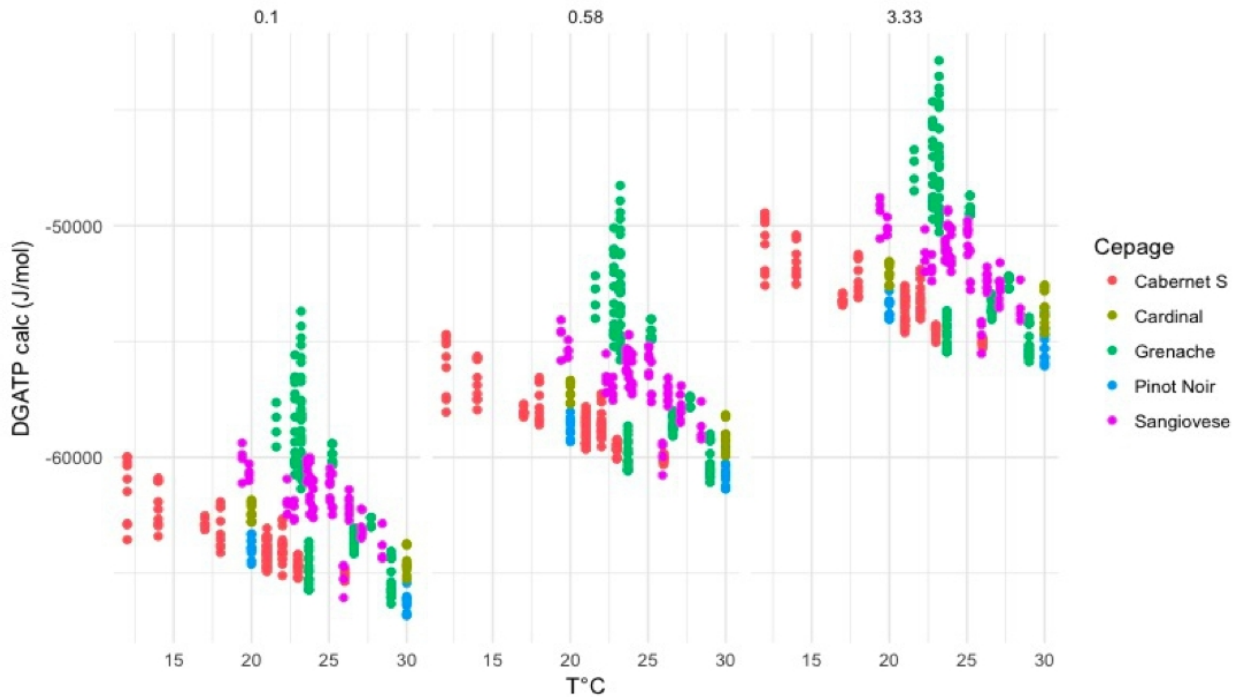
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Annex: supplemental figures.



Supp figure 1. Cytosolic concentration of tartrate computed using measured pH and tartrate using Cabernet Sauvignon dataset. The linear regression is made with “lm” function in R. Coefficients are 0.0027 and -0.0037 (T1 and T2 respectively), $R^2 = 0.878$.



Supp figure 2 : Inversion of Malate model using all the datasets : Variation of $\Delta GATP$ as a function of temperature. $\Delta GATP$ is compute using measured pH and Malate among three values of $[Mal]_{cyt}$: 0.1, 0.58 and 3.33 mM.